Seismic Analysis Platform

Laboratory of Advanced Programming 2025/2026. Team: Lorenzo Pizzi, Antonio Serra, Daniele Falanga. Repository: 2187091_PSF

MISSION AND PROBLEM

Real-time seismic surveillance under unstable geopolitical conditions.

Neutral region constraint: no direct intelligence processing in gateway area and broker area.

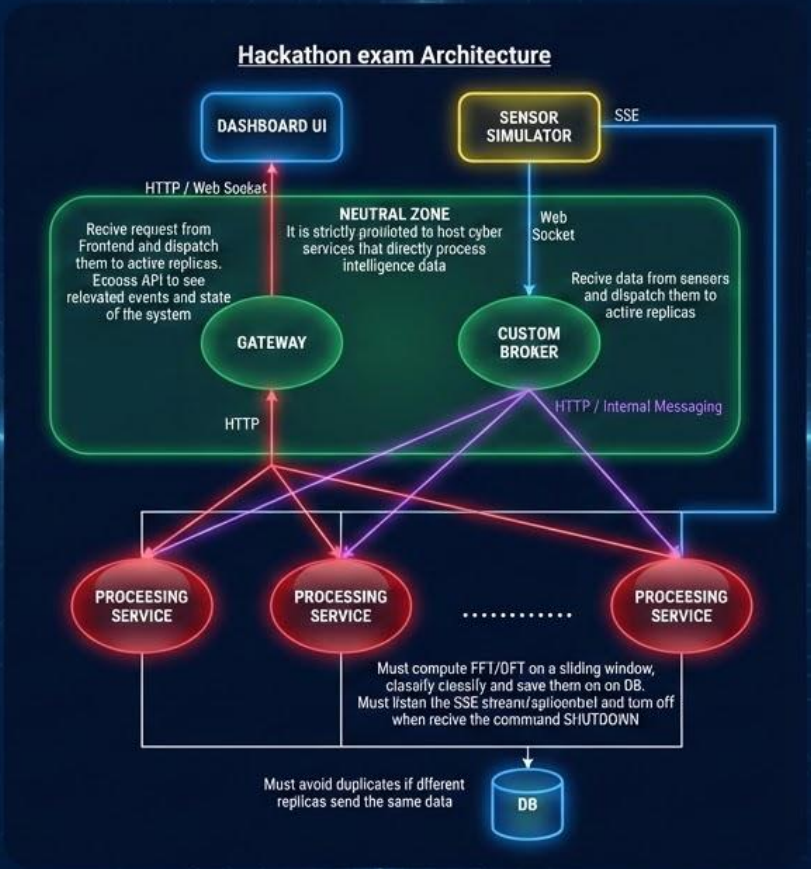
Required outcome: continuous intelligence despite replica failures.

System Goals

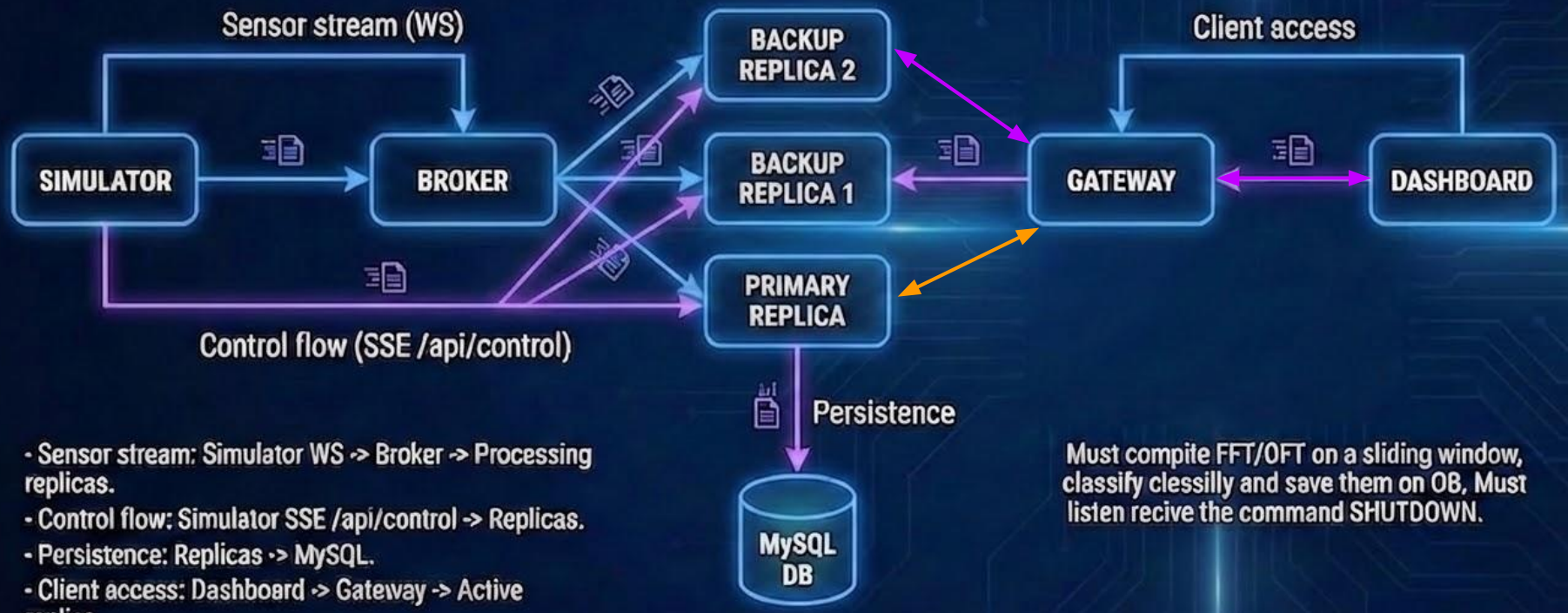
- Ingest real-time simulator data.
- Replicate processing over multiple nodes.
- Perform FFT-based detection and classification.
- Persist events with duplicate-safe behavior.
- Provide real-time and historical observability.

Architecture Overview

- Simulator (provided external container)
- Broker (fan-out)
- Processing replicas (primary + backups)
- MySQL shared database
- Gateway (single entry point + failover)
- Dashboard (live + archive)



Data Flow

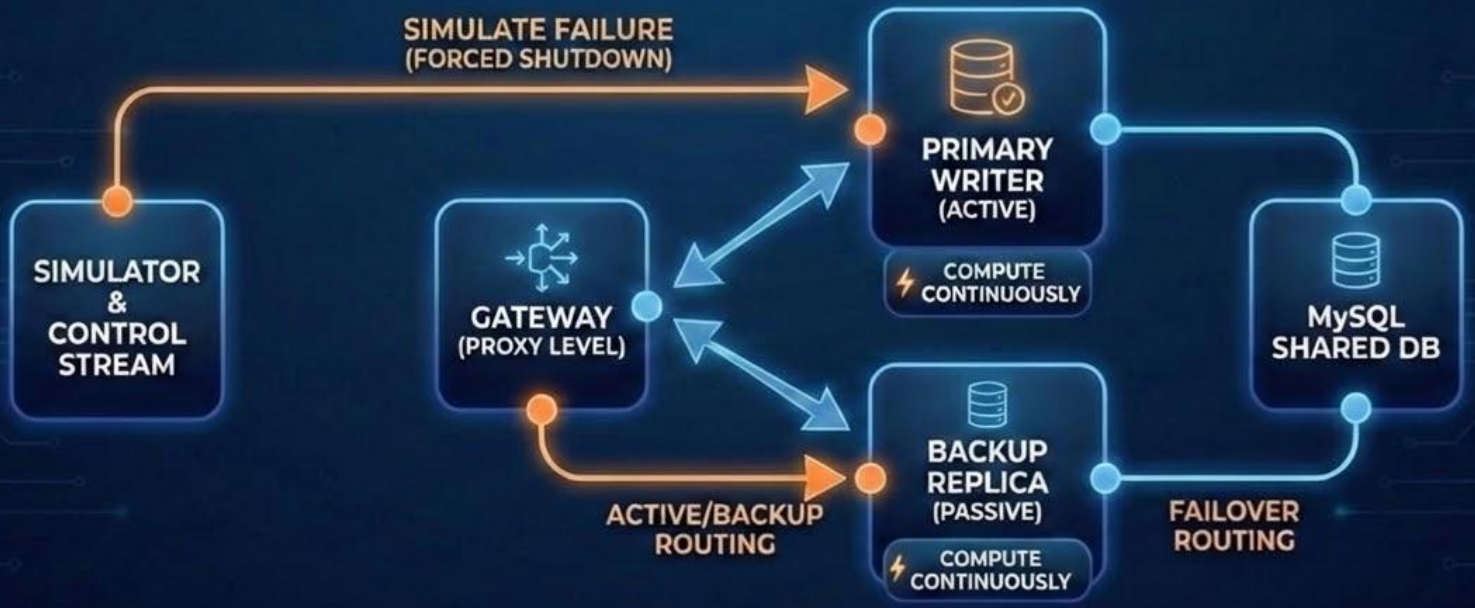


- Sensor stream: Simulator WS -> Broker -> Processing replicas.
- Control flow: Simulator SSE /api/control -> Replicas.
- Persistence: Replicas -> MySQL.
- Client access: Dashboard -> Gateway -> Active replica.

Must compite FFT/OFT on a sliding window, classify clessilly and save them on OB, Must listen recive the command SHUTDOWN.

PROCESSING LOGIC

FAULT TOLERANCE STRATEGY



- Static primary writer (replica-primary).
- Backups compute continuously and persist on primary outage.
- Gateway active/backup routing at proxy level.
- Replica forced shutdown from control stream to simulate failures.

API and User Features

- Health checks and operational counters.
- Event archive endpoint with filters.
- SSE and WS event streaming.
- Dashboard: map, live feed, replica status, archive filters.

Duplicate-Safe Persistence

Deterministic event IDs.

DB lock (GET_LOCK) around critical section.

Time-window dedup query.

INSERT IGNORE fallback for race conditions.

Demo Plan

1. Start stack with Docker Compose.
2. Trigger simulator events manually.
3. Observe live dashboard updates.
4. Trigger shutdown command.
5. Show failover and continued ingestion.
6. Query DB and API to prove persistence and dedup.

NFR Coverage

Availability under partial failure.

Low-latency live feed.

Data consistency and duplicate safety.

Observability through logs and health endpoints.

Reproducible deployment.